



# AbiLearn

## WATER RESOURCES

Geography - Class 10 | Chapter 3 | Deep  
Structured NCERT Notes

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### 1. Introduction to Water Resources

#### Water is Essential for Life

**Water is essential for all forms of life** and is considered the **most precious natural resource**. It is required for **drinking, agriculture, industry, sanitation, and electricity generation**. Without water, **human civilization cannot exist** and **ecosystems would collapse**.

**Water covers about 71% of the Earth's surface**, but **only 2.5% of it is freshwater**. Of this freshwater, **about 68.7% is frozen in glaciers and ice caps**, **30.1% is underground** as groundwater, and **only 0.3% is surface water** in lakes and rivers. This **limited accessible freshwater** must support **all human needs and ecosystems**.

**Water is a renewable resource** but only **if managed sustainably**. The **water cycle** constantly **replenishes freshwater** through **precipitation, evaporation, and condensation**. However, **overuse and pollution** can make water **unavailable for human use**.

## **2. Water Scarcity and Its Causes**

### **What is Water Scarcity?**

**Water scarcity occurs when available water is less than demand** in a region. **According to NCERT**, water scarcity is caused by **population growth, uneven distribution of water, and overexploitation of water resources**. India faces water scarcity despite having **major rivers and adequate rainfall** on average.

**India has 4% of the world's freshwater resources** but supports **18% of the world's population**. This creates **high pressure on limited water resources**. **Per capita water availability in India is declining** due to **population growth and increased consumption**.

**Water scarcity is not just about physical shortage** but also about **lack of access to safe water**. **Polluted water is unusable** even if it is physically present. This is called **quality-based water scarcity**.

## Causes of Water Scarcity (NCERT Points)

- **1. Population Growth:** Rapid population growth is the **main cause of water scarcity** in India. More people need more water for **drinking, cooking, bathing, and sanitation**.
- **2. Expansion of Irrigated Agriculture:** Agriculture is the **largest user of water** in India, using **about 80% of freshwater**. **Water-intensive crops** like **rice and sugarcane** require **large amounts of water**.
- **3. Industrialization:** Industries require **large amounts of water** for **cooling, processing, and cleaning**. **Rapid industrialization** increases demand and also causes **water pollution**.
- **4. Urbanization:** **Urban populations consume more water** than rural populations. **Rapid urbanization** increases water demand in cities, which often depend on **distant water sources**.
- **5. Uneven Distribution of Rainfall:** **Rainfall is not evenly distributed** across India in **space and time**. Some regions receive excess rainfall, while others face drought-like conditions.
- **6. Pollution of Water:** **Industrial waste, domestic sewage, and agricultural chemicals** pollute water bodies. **Contaminated water is unusable**, creating scarcity even where water exists.

### 3. Multi-Purpose River Projects

#### What are Multi-Purpose River Projects?

**Multi-purpose river projects are large dams built across rivers to serve multiple purposes simultaneously. According to NCERT, these projects were called the "temples of modern India" by Jawaharlal Nehru after independence. They were seen as symbols of national development and progress.**

**Dams store river water** and release it according to human needs. They are called multi-purpose because one dam can support **irrigation, power generation, flood control, drinking water supply, navigation, fish breeding, and recreation.**

- **1. Irrigation:** Dams **store water during monsoon** and release it during **dry seasons for irrigation**. This supports **multiple cropping** and improves agricultural production.
- **2. Hydroelectric Power:** Water released from dams **turns turbines** and produces **hydroelectricity**, a **renewable and cleaner energy source**.
- **3. Flood Control:** Dams **store excess monsoon water**, reducing flood intensity downstream.
- **4. Water Supply:** Dams provide **drinking water to cities** and **water for industries**.

- **5. Inland Navigation:** Controlled river flow can make rivers **navigable for transport**.
- **6. Fish Breeding:** Reservoirs provide suitable conditions for **fisheries** and livelihood support.
- **7. Recreation:** Reservoirs support **tourism, boating, fishing, and sightseeing**.

## **Major Multi-Purpose Projects (NCERT Understanding)**

**NCERT focuses on the concept and purposes of multi-purpose river projects.** These projects were promoted after independence because they were expected to support **agriculture, industry, electricity generation, flood control, and national development**.

**Dams transformed many regions** by providing irrigation and electricity, but their benefits were not equal for everyone. They became controversial because of **displacement, environmental damage, and inter-state water conflicts**.

## 4. Criticism of Multi-Purpose Projects

### Why Dams Are Controversial

According to NCERT, multi-purpose river projects have faced criticism for several reasons. **Dams have displaced large numbers of people** without adequate rehabilitation and compensation. **Local communities, especially tribals, have been most affected.**

**Environmental impacts of dams include:**

- **Submergence of forest areas** leading to **loss of biodiversity.**
- **Disruption of river ecosystems** and **fish migration.**
- **Siltation of reservoirs**, which reduces **storage capacity over time.**
- **Reduced downstream flow** and changes in the natural sediment movement of rivers.

**Social impacts include:**

- **Displacement of people** from homes, farms, and traditional habitats.
- **Inadequate rehabilitation** and **poor compensation.**

- **Benefits reaching rich farmers and industries more** than poor communities.
- **Loss of livelihood** for **tribal and rural communities**.

**NCERT also mentions inter-state water disputes.**

States sharing rivers often fight over **water allocation** for **irrigation, drinking water, power generation, and industry**.

## 5. Rainwater Harvesting

### What is Rainwater Harvesting?

**Rainwater harvesting is the collection and storage of rainwater for future use. According to NCERT, it is a traditional practice that is being revived to address water scarcity.** Rainwater harvesting helps in **recharging groundwater and providing water for irrigation and domestic use.**

#### **Benefits of rainwater harvesting:**

- **Recharges groundwater** by allowing rainwater to **percolate into soil**.
- **Reduces flooding** by **slowing down rainwater runoff**.
- **Provides water for irrigation and domestic use.**

- **Reduces dependence on groundwater and surface water.**
- It is **cheap, sustainable, and environmentally friendly.**

## **Traditional Rainwater Harvesting Methods in India (NCERT Content)**

- **1. Guls and Kuls (Western Himalayas): Simple diversion channels** that **direct water from streams** to agricultural fields for **irrigation.**
- **2. Kund (Rajasthan): Underground tanks** that **collect rainwater from rooftops** and store it for drinking during dry months.
- **3. Tankas (Rajasthan): Underground tanks** attached to **house rooftops**, especially in **Bikaner, Phalodi, and Barmer**, used for drinking water during summer.
- **4. Bamboo Drip Irrigation (Meghalaya): Bamboo pipes** carry water from springs to fields using **gravity flow.**
- **5. Inundation Channels (Bengal Floodplains):** Channels carry **flood water** from rivers to fields for **irrigation.**
- **6. Rooftop Rainwater Harvesting (Rajasthan):** Rainwater from roofs is collected and stored in



**underground tankas** for future use.

## Why Rainwater Harvesting is Important

- **1. Recharges Groundwater:** Rainwater harvesting **raises groundwater levels** and improves future water availability.
- **2. Reduces Water Scarcity:** Stored rainwater reduces dependence on groundwater and distant water sources.
- **3. Reduces Flooding:** It slows runoff and lowers flood pressure during heavy rain.
- **4. Sustainable and Cheap:** It uses local materials and needs less energy.
- **5. Protects Traditional Knowledge:** It revives water management methods used by Indian communities for centuries.

## 6. Modern Rainwater Harvesting

### Rooftop Rainwater Harvesting

**Modern rooftop rainwater harvesting involves collecting rainwater from rooftops and storing it in tanks or recharging groundwater. NCERT mentions that rooftop rainwater harvesting is becoming popular in many parts of India.**

**How rooftop rainwater harvesting works:**

- **Rainwater from rooftops** is collected through **pipes**.
- Water is **filtered** to remove **debris and contaminants**.
- Water is **stored in tanks** above ground or underground.
- Excess water is directed to **recharge pits** or **bore wells** to **recharge groundwater**.

**NCERT mentions that Chennai made rooftop rainwater harvesting mandatory** for all buildings. This helped **raise groundwater levels** in the city.

## 7. Water Conservation

### Why Water Conservation is Important

**According to NCERT, water conservation is essential** because **water is finite** and **demand is increasing**.

**Without conservation, water scarcity will worsen** in the future. Water conservation ensures **water availability for future generations**.

#### **Benefits of water conservation:**

- **Ensures water availability** for future generations.
- **Reduces cost** of water treatment and supply.
- **Protects ecosystems** that depend on water.

- **Reduces energy consumption** because pumping and treating water require energy.
- **Prevents conflicts** over water resources.

## Methods of Water Conservation

- **1. Efficient Irrigation:** Using **drip and sprinkler irrigation** instead of **flood irrigation** saves **30-70% water**.
- **2. Fixing Leaks:** **Fixing leaking taps and pipes** prevents water wastage.
- **3. Reducing Consumption:** **Shorter showers, turning off taps while brushing, and full loads in washing machines** save water.
- **4. Recycling Water:** **Treating and reusing wastewater** reduces demand for fresh water.
- **5. Rainwater Harvesting:** **Collecting and storing rainwater** reduces dependence on other water sources and improves groundwater recharge.

## 8. Key NCERT Points Summary

### Important Facts from NCERT

- **1. Water Distribution:** **71% of Earth's surface** is covered with water, **only 2.5% is freshwater**, and **less than 1% is easily accessible**.

- **2. India's Water Position:** India has **4% of the world's freshwater** but supports **18% of the world's population**.
- **3. Multi-Purpose Projects:** Dams were called "**temples of modern India**" and serve **irrigation, power, flood control, water supply, navigation, and fish breeding**.
- **4. Traditional Rainwater Harvesting:** Important methods include **guls and kuls, kunds and tankas, bamboo drip irrigation, and inundation channels**.
- **5. Rainwater Harvesting Benefits:** It **recharges groundwater, reduces water scarcity, reduces flooding, and is sustainable and cheap**.